1)

a. $\int \frac{x^2}{\sqrt{1-x}}\, dx$

let $u = 1-x$

$du = -dx$

$x = 1-u$

$x^2 = (1-u)^2$

$\int \frac{1-u^2}{\sqrt{u}}(-du) = \int (1-u)^2 u^{-\frac{1}{2}} du$

$(1-u)^2 = 1-2u+u^2$

$(1-2u+u^2)\, u^{-\frac{1}{2}}$

$\int (u^{-\frac{1}{2}} - 2u^{\frac{1}{2}} + u^{\frac{3}{2}})$

$\frac{2u^{\frac{1}{2}}}{2} + \frac{4}{3}u^{3/2} + \frac{2}{5}u^{5/2}$

b. $\int (\ln x)^3 \, dx$

c. $\int x^{11} e^{x^6} \, dx$

Area curve $x = y^2 - 4y$

$x = 2y - y^2$

$y^2 - 4y = 2y - y^2$

$2y^2 - 6y = 0$

$2y(y-3) = 0$

$y = 0 \qquad y = 3$

test on $0, 3$

$y = 1$

$x = -3$

$x = 1$

Shell method

$$y = x^2$$

$$y = 0$$

$$x = 1$$

$$x = \sqrt{y} \qquad 1 = \sqrt{y} \qquad 1 = y$$

$$0 = y$$

right: $x = 1$

left: $x = \sqrt{y}$

$$V = 2\pi \int \text{radius} \cdot \text{height}\, dy$$

$$r = y$$

$$\text{height} = 1 - \sqrt{y}$$

$$V = 2\pi \int_0^1 y(1 - \sqrt{y})\, dy$$

disk method